

Interim Report

Project Title

Spill: A Mobile Social and Mental **Wellness** Application

1. Introduction

1.a Problem / Question / Hypothesis

The project addresses the following:

Problem: Many people lack a safe, low-friction way to share **mental health** experiences, track **mood**, and connect with others without the permanence and pressure of traditional **social media**. Existing apps often focus either on clinical **therapy** or on generic social feeds, leaving a gap for a community-oriented, **wellness**-focused space.

Question: Can a mobile application combine (1) anonymous or pseudonymous sharing with time-limited "**vent**" posts, (2) **mood** and **gratitude** tracking, (3) direct messaging and match-based connection, and (4) optional **premium** features, delivered via a cross-platform (iOS/Android) app with a modern tech stack, to support users' **mental wellbeing** and sense of community?

Hypothesis: A single, well-designed mobile app that integrates sharing (including ephemeral **vent** posts), **mood/gratitude** logging, connections, and **premium** features will be technically feasible, usable, and aligned with user needs for a supportive, non-clinical **mental wellness** space.

1.b Importance, Significance and Novelty

Importance: **Mental health** remains a major concern; accessible, non-stigmatising tools that encourage expression and connection can complement formal care. A mobile-first approach reaches users where they are.

Significance: The project demonstrates end-to-end **software development** (requirements, design, implementation, database, auth, real-time features) and addresses a real-world domain with social and **wellness** impact.

Novelty: The combination of (1) **vent** mode (time-limited, disappearing posts) for low-commitment sharing, (2) **mood** and **gratitude** tracking with optional AI-assisted insights, (3) matches based on compatibility and (4) **premium** membership within one cross-platform app, using an open-source stack (**Expo/React Native, Supabase**), offers a distinct product. Using **Supabase** and **Expo** provides a modern, cost-effective alternative to fully proprietary backends and native-only development.

2. Background / Literature Review

2.1 Domain and Related Work

Mental health apps: Numerous apps exist for **mood** tracking, meditation, or **therapy** (e.g. **Headspace, Calm, Woebot**). Many are either clinical/self-help focused or generic **social networks**. This project sits in the middle: community and sharing with **wellness**-oriented features (**mood, gratitude, vent** mode).

Ephemeral content: Time-limited content (e.g. **Stories**) is well established in social products. Applying a "**vent** mode" to **mental health** sharing reduces perceived permanence and can encourage honest expression.

Social support and peer connection: Research indicates that peer support and feeling connected can benefit **mental wellbeing**. The project's connections, matches, and feed are designed to support that.

2.2 Underpinning Technology

Cross-platform mobile: **React Native (Expo)** enables one codebase for iOS, Android, and web, with native performance and access to device APIs (camera, sensors, haptics).

Backend and data: **Supabase** provides **PostgreSQL**, real-time subscriptions, authentication (email/password, magic link), and storage (images/videos), reducing custom backend work.

Front-end stack: **Expo Router** for file-based navigation, **NativeWind** (Tailwind-style styling), **Reanimated** for animations, and **TypeScript** for type safety and maintainability.

3. Aim and Objectives

3.a Aim

To design, implement, and evaluate a cross-platform mobile application (**Spill**) that supports users' **mental wellness** through:

Safe, category-based sharing (including time-limited **vent** posts),

Mood and **gratitude** tracking with optional AI-assisted insights,

Connections (messaging and matches),

Optional **premium** membership and a clear path to future monetisation.

3.b Objectives and Tasks

1. User authentication and profiles. Implement sign-up, login (including magic link), password reset, and profile management (display name, avatar, anonymous username).
2. Feed and posts. Implement a feed with categories (e.g. **General, Anxiety Share, Depression Vent**), upvote/downvote, comments, and media (image/video). Implement **vent** mode: time-limited posts (e.g. 24 hours) that disappear after expiry.
3. **Mood** and **gratitude**. Allow users to log **mood** (e.g. 1 to 5 scale) and **gratitude** entries. Provide visualisations (e.g. **mood** over time) and, where applicable, AI-generated **gratitude** suggestions (**premium**).
4. Connections and messaging. Implement direct messaging (conversations, real-time updates). Implement a Matches feature (e.g. compatibility or activity-based) with optional **premium** gating.
5. **Premium** membership. Define and implement **premium** features (e.g. **vent** mode, AI **gratitude**, matches, ad-free experience). Implement **premium** status checks, activation, and cancellation flows and a payment/**premium** welcome flow.
6. Additional features. Notifications (e.g. new followers, messages), groups, streaks (e.g. **mood** or activity streaks), and settings (privacy, account, **premium**).
7. Technical quality. Use a structured backend (**Supabase**: tables, RLS, storage, real-time). Ensure the app is responsive, accessible, and maintainable (**TypeScript**, clear structure).

3.c Target Audience

Primary: Adults (e.g. 18 to 35) seeking a non-clinical space to share experiences, track **mood** and **gratitude**, and connect with others.

Secondary: Anyone looking for a lightweight **mental wellness** and community app on iOS or Android.

Deliverables: A working **Expo/React Native** application backed by **Supabase**, deployable to app stores; database migrations and documentation; and a final report with evaluation.

4. Methodology

4.1 Methodological Approach

Approach: **Agile**-inspired iterative development: define core features (auth, feed, **mood**, connections, **premium**), implement in phases, and test continuously.

Rationale: Requirements in social/**wellness** apps benefit from iteration; **Supabase** and **Expo** allow fast prototyping and changes.

Alternatives considered: Native (**Swift/Kotlin**) or **Flutter** would increase platform-specific effort; a custom backend would extend timeline. The chosen stack balances speed, maintainability, and cross-platform reach.

4.2 Analysis and Specification of Requirements

Functional requirements (summary):

Authentication: Register, login (password and magic link), logout, password reset; session persistence and secure storage (e.g. **AsyncStorage** on device).

Profiles: Create/update profile (display name, avatar, anonymous username); view other users' profiles where applicable.

Posts: Create post (text, optional image/video, category); **vent** mode (duration, expiry); view feed by category; upvote/downvote; comment; delete own post; filter expired **vent** posts.

Mood and **gratitude**: Log **mood** (1 to 5) with optional note; log **gratitude**; view history and averages; optional AI **gratitude** suggestion (**premium**).

Connections: List conversations; send/receive messages; real-time updates; unread counts.

Matches: View compatible users; send match request (**premium**); optional voice/audio and weekly summary (**mood** trend, insights).

Premium: Check status; activate/cancel **premium**; gate **vent** mode, AI features, match requests, etc.

Other: Notifications, groups, streaks, settings (account, privacy, **premium**).

Non-functional requirements:

Usable on iOS and Android (and optionally web).

Responsive UI and smooth interactions (e.g. **Reanimated**).

Data secured via **Supabase** (HTTPS, RLS, auth).

Real-time where needed (messages, notifications).

4.3 Design

Architecture: Client (**Expo/React Native** app) and **Supabase** (Auth, **PostgreSQL**, Realtime, Storage).

Navigation: File-based routing (**Expo Router**): tabs (Home, **Mood**, Connections, Matches), and stacks for auth, post creation, profile, settings, etc.

Data model (high level): profiles (user id, display_name, avatar_url, anonymous_username, is_**premium**, **premium_activated_at**, **premium_expires_at**). *posts* (id, user_id, content, category, media_url, isvent, expires_at, created_at). post_votes, post_stats, comments; **mood**_entries, **gratitude**_entries; conversations, messages; followers; matches / match_requests; streaks, groups, etc.

Security: Row Level Security (RLS) so users access only their or permitted data; auth tokens and secure storage for sessions.

5. Implementation

5.1 Progress to Date

Implementation has been carried out in line with the methodology and design.

5.1.1 Technology stack

Front-end: **Expo** SDK 54, **React Native**, **TypeScript**, **Expo Router**, **NativeWind** (Tailwind), **React Native Reanimated**, expo-image, expo-video, expo-image-picker, expo-sensors (e.g. accelerometer for shake-to-**gratitude**), Lottie, react-native-chart-kit.

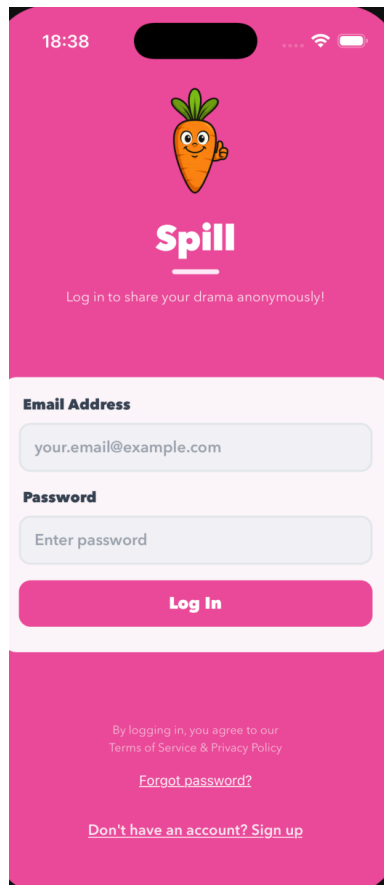
Backend and services: **Supabase** (Auth, **PostgreSQL**, Realtime, Storage buckets for images and video).

State and data: **Supabase** client with **AsyncStorage** for session persistence; real-time channels for messages and notifications.

5.1.2 Authentication and profiles

Implemented: sign-up, login (email/password and magic link), logout, forget password, reset password, auth callback handling.

Profile fields (display_name, anonymous_username, avatar_url) and **premium** fields (is_**premium**, **premium_activated_at**, **premium_expires_at**) are stored in profiles and used across the app.



Spill login screen (email, password, magic link).

5.1.3 Feed and posts

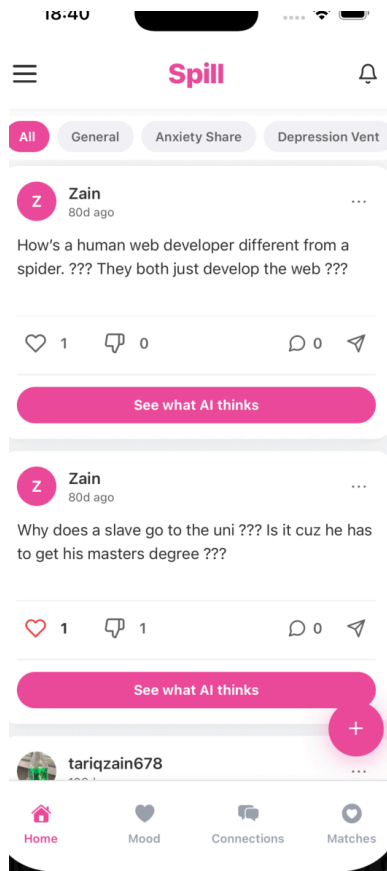
Home feed with categories: All, **General**, **Anxiety Share**, **Depression Vent**.

Create post: text, category, optional image/video upload to **Supabase** Storage; **vent** mode with configurable duration (e.g. 24 h) and expires_at.

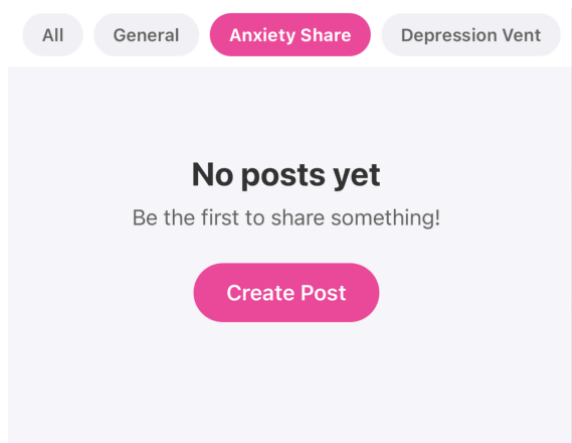
Feed filters out expired **vent** posts; **vent** posts show a **VENT** badge and remaining time.

Upvote/downvote, comments, view counts, delete post, and AI opinion on posts implemented.

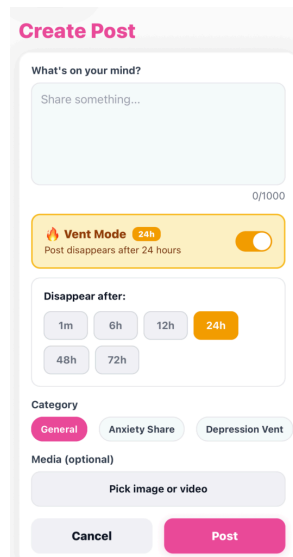
Migrations: vent-mode-migration.sql adds is_vent, expires_at, indexes, and optional cleanup function for expired vents.



category tabs (All, General, Anxiety Share, Depression Vent)



Home feed with selected categories but it shows none because we don't have any post in this.



Create post screen with **vent** mode (time-limited post) option.

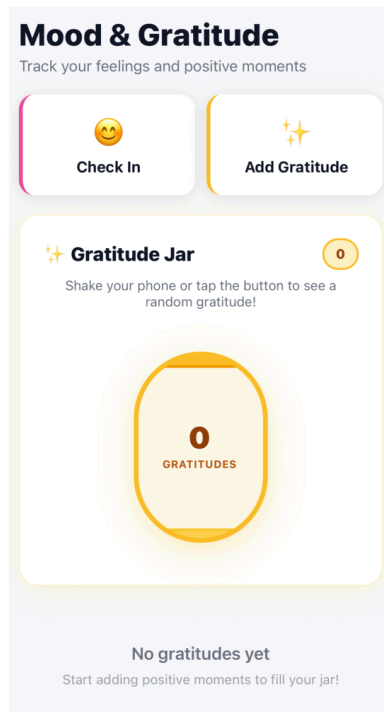
5.1.4 Mood and gratitude

Mood tab: log **mood** (1 to 5 with labels/emojis), optional note; list **mood** entries; average **mood**; **mood** chart.

Gratitude: add **gratitude** entries; view count; shake to get random **gratitude** (accelerometer); optional AI-generated **gratitude** suggestion (**premium**).

Data in **mood_entries** and **gratitude**-related tables; functions such as **logMood**, **getMoodEntries**, **getAverageMood**, **getGratitudeEntries**, **generateAIGratitude**.

Mood tab



Mood and **gratitude** tab (**mood** logging and **gratitude** entries).(stiull in testing)

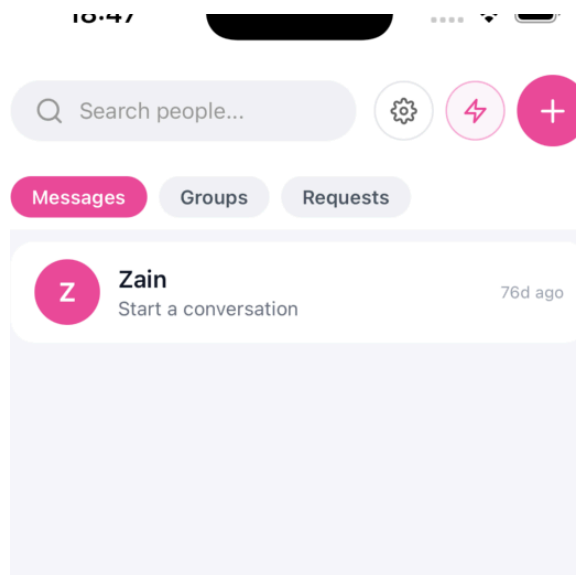
5.1.5 Connections and messaging

Connections tab: list conversations, real-time messages, unread message count in tab badge.

Realtime subscriptions for new messages and new followers.

Navigation to messaging screen and integration with **Supabase** conversations and messages.

Connections tab :



Connections tab (conversations and messaging).

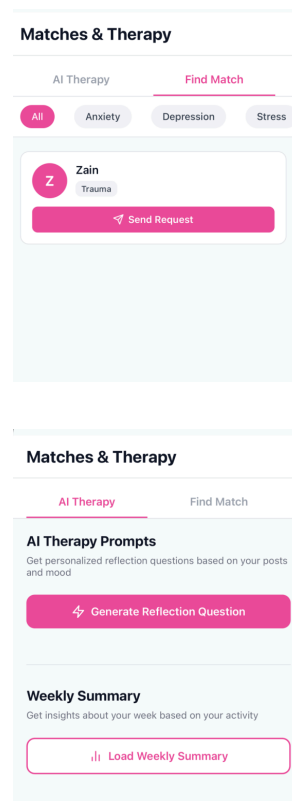
5.1.6 Matches

Matches screen: view other users (e.g. by compatibility/**mood**); send match request (**premium**-gated).

Weekly summary with **mood** trend (improving/stable/declining) and AI-generated insights.

Realtime channels for match-related events (e.g. voice status, audio-ready).

Migration: matches-migration.sql for match-related schema.



Matches screen (match requests and weekly summary)

5.1.7 Premium

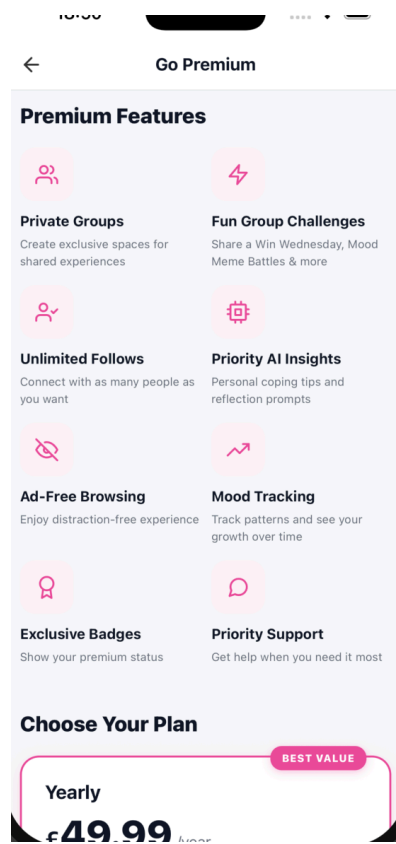
Premium status checked via `checkPremiumStatus()` (`profiles.is_premium`).

Activate/cancel **premium** implemented (`activatePremium`, `cancelPremium`).

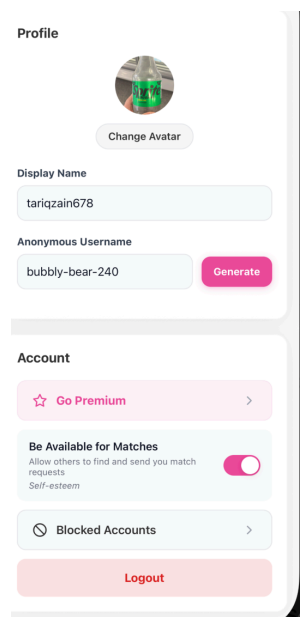
Premium welcome and payment flows (screens: **premium**, **premium-welcome**, **payment**, **success**).

Premium gating applied to **vent** mode, AI **gratitude**, match requests, and reflected in UI (e.g. Go **Premium** prompts).

Migration: **premium**-migration.sql for is_premium, premium_activated_at, premium_expires_at.



Premium membership screen.(will outsource this service via a stripe or any other vendor . still testing.



Settings screen.

5.1.8 Additional features

Notifications: Real-time follower and message counts; tab badges for unread.

Groups and streaks: Group creation/joining; streak types (e.g. activities); check-in; leaderboards; getUserStreaks, getAllUserStreaks, checkInToStreak, etc.

Profile and settings: Profile screen; settings (account, privacy, **premium** management, logout).

UI/UX: Hamburger menu on home, pull-to-refresh, haptics, themed components, error and loading states.

5.1.9 Backend and data

Supabase project with tables for profiles, posts, post_votes, post_stats, comments, **mood**_entries, **gratitude**, conversations, messages, followers, matches, streaks, groups, etc.

Storage buckets for images and video.

Centralised logic in app/functions.tsx (fetchPosts, createPost, vote, comment, **mood**, **gratitude**, streaks, **premium**, AI helpers, weekly summary).

5.2 Summary of Implementation Status

Auth and profiles: Done. Email/password, magic link, reset, profile fields.

Feed and posts: Done. Categories, **vent** mode, media, votes, comments.

Mood and **gratitude**: Done. Log, history, chart, AI suggestion (**premium**).

Connections: Done. Conversations, real-time messages, badges.

Matches: Done. Match requests (**premium**), weekly summary.

Premium: Done. Status, activate/cancel, gating.

Groups and streaks: Done. Create/join, check-in, leaderboard.

Notifications: Done. Realtime counts and badges.

Settings: Done. Account, privacy, **premium**.

6. Testing and Evaluation Strategy

6.1 Success Criteria

Functional: All major user flows work (register, login, view feed, create post, **vent** mode, **mood** log, **gratitude**, messages, matches, **premium**).

Technical: No critical crashes; real-time updates work; expired **vent** posts are hidden; **premium** gating enforced.

Usability: Key tasks completable without documentation; UI consistent and responsive.

6.2 Testing Approach

Unit and logic: Critical logic (e.g. **mood** average, streak calculation, **vent** expiry filter) can be tested in isolation.

Integration: Test **Supabase** operations (auth, CRUD, realtime) against a development project.

Manual and UI: Test on iOS and Android (simulator/device) for navigation, forms, media upload, and real-time behaviour.

User evaluation (planned): Short usability tests or surveys with target users to assess usefulness, ease of use, and perceived safety of **vent** mode and sharing.

6.3 Evaluation of Results

Compare implemented features against the requirements and objectives.

Report on any limitations (e.g. payment integration is placeholder/demo if not integrated with a real provider).

Document user feedback and reflections in the final report.

7. Professional, Social, Economic and Legal Issues

Professional: Follow good practice in **software engineering** (version control, structure, documentation). Do not provide clinical or therapeutic advice; the app is a supportive community and self-tracking tool, not a replacement for professional care.

Social: Design for respectful, inclusive use; consider moderation and reporting for user-generated content; emphasise privacy (anonymous/pseudonymous options, **vent** mode).

Economic: **Premium** model supports sustainability; use of open-source (**Expo**, **Supabase**) keeps development and running costs manageable.

Legal: Comply with data protection (e.g. **GDPR**/UK **GDPR**) for personal data; ensure terms of service and privacy policy; secure handling of auth and sensitive data (**Supabase** best practices, RLS).

8. Resource Requirements

Development: IDE (e.g. **VS Code**/Cursor), **Node.js**, **Expo** CLI, iOS Simulator / **Xcode**

Backend: **Supabase** (free tier sufficient for development and small-scale evaluation).

Optional: **OpenAI** (or similar) API for AI features if used in production; payment provider for real **premium** monetisation.

9. References

1. **Expo** Documentation. (2024). **Expo** SDK Documentation. <https://docs.expo.dev/>
2. **Supabase**. (2024). **Supabase** Documentation. <https://supabase.com/docs>
3. **React Native**. (2024). **React Native** Documentation. <https://reactnative.dev/docs/getting-started>
4. Meta. (2024). **React** Documentation. <https://react.dev/>
5. **NativeWind**. (2024). **NativeWind** – Tailwind for **React Native**. <https://www.nativewind.dev/>

Appendix 1 – Project timeline (Outline)

Proposal and requirements: Week 1 to 2

Design (data model, UI): Week 3 to 4

Auth and profiles: Week 4 to 5

Feed, posts, **vent** mode: Week 5 to 7

Mood and **gratitude**: Week 7 to 8

Connections and messaging: Week 8 to 9

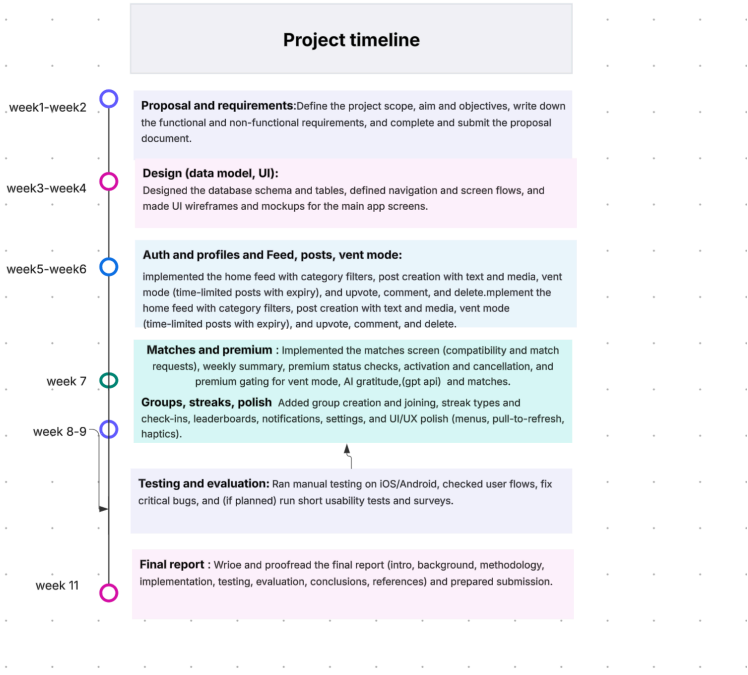
Matches and **premium**: Week 9 to 10

Groups, streaks, polish: Week 10 to 11

Testing and evaluation: Week 11

Final report: week 11

Gantt chart



Appendices (Where Appropriate)

Appendix A: Database schema diagrams (from Supabase).



This interim report follows the project/dissertation interim report structure provided in the guideline and reflects progress to date on the **Spill** project. It should be read as a continuation of the proposal, with added content showing current progress and a clear plan for completion.